

COMSATS UNIVERSITY ISLAMABAD



SRS Document

for

Job Portal

By

Noman Sharif CIIT/SP24-BAi-041/ISB

To

Sir Aamir Shabbir Parre

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1. Introduction:

1.1. Purpose:

This document outlines the requirements for the **Job Portal (JP)**. Much like a digital marketplace, this system connects people looking for work with companies looking for talent.

1.2. Scope:

The Job Portal will:

- Allow Job Seekers to:
 - Create a professional profile and upload resumes.
 - Search for jobs using filters like location and salary.
 - Track the status of their applications in real-time.
- Allow Recruiters to:
 - Post new job openings and manage candidates.
 - Use automated tools to find the best-matched talent.
- Allow Administrators to:
 - Oversee the platform, verify companies, and manage all users.

2. Overall System Description:

2.1. System Architecture:

The system is built in three layers to ensure it is fast and reliable:

- **The Interface (Client Layer):** What users see on their screens, built with Next.js for high speed and Google-search visibility.
- **The Brain (Application Layer):** A Node.js backend that handles the "logic," such as matching resumes to jobs.
- **The Memory (Database Layer):** A PostgreSQL database that securely stores all user information.

2.2. User Roles:

Role	Responsibility
Job Seeker	Builds a profile, uploads a resume, and applies for positions.
Recruiter	Creates job posts and chooses the best candidates.
Admin	Manages the system settings, users, and security.

3. Functional Requirements:

3.1. Job Seeker Module:

3.1.1. Registration & Authentication

FR-U1: User Registration

- User shall register with:
 - Full Name
 - Email
 - Phone Number
 - Password
- System shall validate inputs.
- Password must be hashed (e.g., bcrypt).

FR-U2: User Login

- User shall login using email/phone and password.
- System shall generate session/token (JWT recommended).

3.1.2. Profile & Resume Management

FR-U3: Create/Edit Profile

- User shall enter:
 - Professional Summary
 - Education History
 - Work Experience
 - Skills List
- User shall upload Resume (PDF/Docx).
- System shall store file metadata in PostgreSQL.

3.1.3. Job Search & Filtering

FR-U4: Search for Jobs

- User shall search by:
 - Keywords (Job Title/Skills)

- Location
 - Job Category (Remote/On-site)
- System shall validate search inputs.
- System shall display results based on relevance.

3.1.4. Job Application

FR-U5: Apply for Job

- User shall:
 - Select a Job Post
 - Attach saved Resume
 - Submit Application
- System shall:
 - Store application as "Applied".
 - Calculate "Match Score" using the search algorithm.

3.1.5. Application History

FR-U6: View Application History

- User shall view:
 - Application ID
 - Job Title
 - Company Name
 - Date Applied
 - Status (Applied, Shortlisted, Rejected)

3.2. Recruiter Module:

3.2.1. Recruiter Authentication

- Recruiter shall login using assigned credentials.
- Recruiter shall view their posted job listings.

3.2.2. Job Management

FR-R1: Create Job Posting

- Recruiter shall enter:
 - Job Title
 - Detailed Description
 - Required Skills
 - Salary Range
- System shall store posting as "Open".

3.2.3. Candidate Management

FR-R2: View Applicants & Match Scores

- Recruiter shall:
 - View list of applicants for a specific job.
 - View "Match Score" for each candidate.
- System shall:
 - Rank candidates using the search algorithm.
 - Display candidate profile details.

FR-R3: Update Application Status

- Recruiter shall:
 - Change status to "Shortlisted" or "Rejected".
- System shall:
 - Update database status.
 - Notify the Job Seeker of the status change.

3.3. Admin Module:

3.3.1. User & Recruiter Management

FR-A1: Manage Users

- Admin shall:
 - View all Seeker and Recruiter details.

- Activate/Deactivate accounts.

3.3.2. Job Moderation

FR-A2: Moderate Job Postings

- Admin shall:
 - Review new job listings.
 - Approve or Delete inappropriate postings.

3.3.3. System Overview

FR-A3: View Platform Analytics

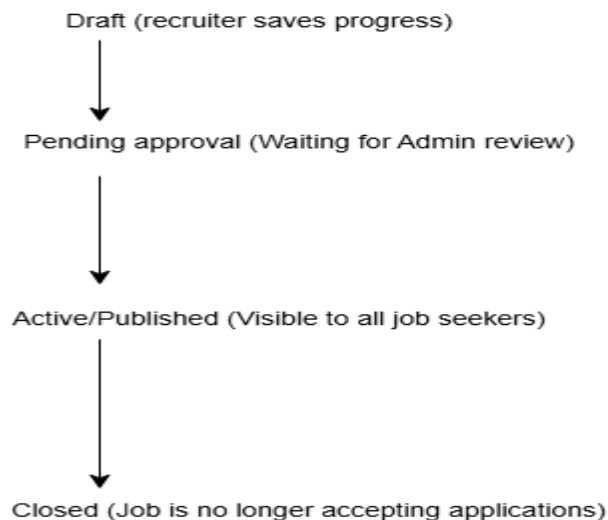
- Admin shall filter data by:
 - Date
 - User Type
 - Job Status

4. Job & Application Status Lifecycle

This section defines how a job posting and a candidate's application move through the system, ensuring all parties (Seeker, Recruiter, and Admin) are aligned on the current stage of the process.

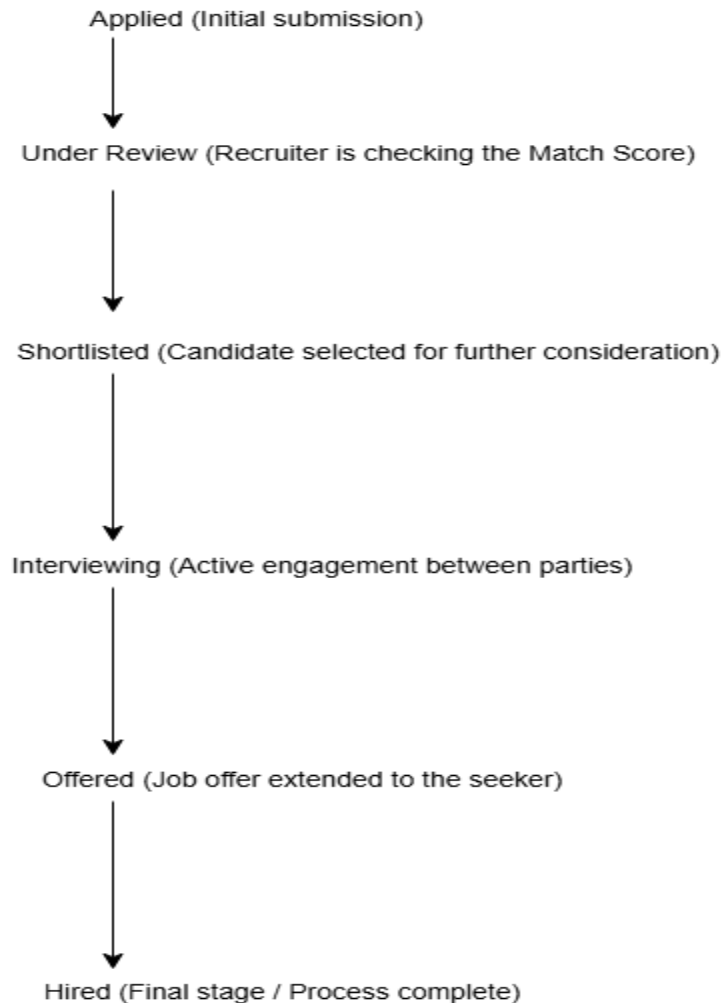
4.1. Job Posting Lifecycle (Recruiter Perspective)

This flow tracks a job advertisement from creation to being filled.



4.2. Application Status Lifecycle (Seeker Perspective)

This flow tracks a specific seeker's journey after they click "Apply."



4.3. Alternative Transitions

A job process may end prematurely:

- **Withdrawn by Seeker:** The candidate chooses to remove their application.
- **Rejected by Recruiter:** The candidate does not meet the requirements for the role.
- **Deleted/Flagged by Admin:** The job post is removed due to a violation of platform rules.
- **Position Cancelled:** The recruiter closes the role without hiring anyone.

5. Database Requirements (High-Level Schema)

5.1. Users Table

This table stores the core credentials and identity information for all platform members (Seekers, Recruiters, and Admins).

Field	Type
id	PK
name	VARCHAR
email	VARCHAR (Unique)
phone	VARCHAR
password_hash	TEXT
role	ENUM (Seeker, Recruiter, Admin)
created_at	TIMESTAMP

5.2. Jobs Table

This table acts as the "inventory" of the system, similar to the Cabs table in the reference document.

Field	Type
id	PK
recruiter_id	FK
title	VARCHAR
description	TEXT
location	VARCHAR
salary_range	VARCHAR
status	ENUM (Open, Closed, Draft)
created_at	TIMESTAMP

5.3. Applications Table

This table functions as the "Booking" log, tracking which seeker has applied for which job.

Field	Type
id	PK
job_id	FK
seeker_id	FK
resume_url	TEXT
match_score	INT
status	ENUM (Applied, Shortlisted, Rejected, Hired)
applied_at	TIMESTAMP

5.4. Skills Table

This table defines the standardized skills used for matching seekers to job postings.

Field	Type
id	PK
skill_name	VARCHAR (Unique)

5.5. Job Skills Table

This table functions similarly to the **Via Routes** table, linking multiple requirements to a single job posting.

Field	Type
id	PK
job_id	FK
skill_id	FK
importance_weight	INT

6. Non-Functional Requirements

6.1. Performance

- **Concurrent Users:** System must support 1000+ concurrent users without degradation in performance.
- **Response Time:** API response time for job searches and profile loading must be < 2 seconds.
- **Search Optimization:** The search algorithm must return matched results in < 500ms for a database of 50,000+ records.

6.2. Security

- **Data Protection:** Password hashing using **bcrypt** is mandatory.
- **Authentication:** Implementation of **JWT-based authentication** for secure sessions.
- **Access Control: Role-based access control (RBAC)** to ensure Seekers, Recruiters, and Admins only access their respective modules.
- **Integrity:** Strict input validation and sanitization to prevent SQL injection and XSS attacks.
- **Encryption:** HTTPS protocol is mandatory for all data in transit.
- **File Security:** Resume uploads must be scanned and stored in a secure, isolated storage bucket.

6.3. Availability

- **Uptime:** 99% uptime target to ensure the platform is accessible to global recruiters.
- **Backup:** Daily automated database backups to prevent data loss.

6.4. Scalability

- **Architecture:** Modular architecture to allow independent updates to the Seeker or Recruiter modules.
- **API Design:** API-first design to support future mobile application integration.
- **Cloud Ready:** Fully deployable on cloud infrastructure such as **AWS, Azure, or GCP**.

6.5. Usability

- **Responsiveness:** Mobile responsive UI developed with Next.js to support users on all devices.

- **Workflow:** Clear job application flow, requiring a maximum of 3 steps from discovery to submission.
- **SEO:** Frontend must be optimized for search engines to ensure job postings appear in external search results.

7. Business Rules

1. **Unique Applications:** A job seeker cannot apply for the same job posting more than once.
2. **Resume Requirement:** A user must have a resume uploaded to their profile before the "Apply" button becomes active.
3. **Job Expiration:** Job postings will automatically move to "Closed" status after 30 days unless manually renewed by the recruiter.
4. **Admin Moderation:** Only an Administrator has the authority to permanently delete job postings or user accounts from the database.
5. **Recruiter Verification:** A recruiter must be marked as "Active" or "Verified" by an Admin to publish job listings to the public feed.
6. **Application Withdrawal:** A seeker may withdraw their application only if the status is "Applied" or "Under Review." Once "Interviewing" begins, withdrawal requires manual coordination.
7. **Data Integrity:** Job descriptions must meet a minimum character count to ensure quality listings for seekers.

8. Future Enhancements

These features are planned for subsequent versions of the platform to improve the user experience and matching efficiency:

- **AI Resume Parsing:** Implementation of Natural Language Processing (NLP) to automatically extract skills and experience from uploaded PDF resumes to populate user profiles.
- **Real-time Chat System:** Integration of WebSockets to allow instant messaging between Recruiters and Seekers for interview scheduling.
- **Video Interview Module:** Built-in video conferencing tool to conduct initial screening rounds directly within the portal.

- **Premium Job Listings:** Online payment integration (e.g., Stripe/PayPal) to allow companies to "Feature" their job posts at the top of search results.
- **Company Reviews:** A rating and feedback system where former employees can rate company culture and hiring processes.
- **Map-based Job Search:** Integration of a Map API (e.g., Google Maps) to allow seekers to visualize job locations and commute distances.
- **LinkedIn/GitHub Integration:** One-click profile import from professional social networks to speed up the registration process.

9. Acceptance Criteria

The system will be considered complete and ready for deployment when it satisfies the following conditions:

- **Functional Completion:** All functional requirements (FR-U1 through FR-A3) are fully implemented and operational.
- **Role Validation:** Successful end-to-end testing has been performed for all three roles: Job Seeker, Recruiter, and Administrator.
- **Data Integrity:** Database referential integrity is maintained, ensuring no "orphan" applications exist without a corresponding job or user.
- **Security Audit:** No critical security vulnerabilities are present, and JWT authentication correctly restricts access based on roles.
- **User Success Path:** A Job Seeker can successfully:
 - Register and upload a resume.
 - Search and apply for a job.
 - View their application status history.
- **Recruiter Success Path:** A Recruiter can successfully:
 - Post a job and see it appear in the public feed.
 - View a ranked list of applicants based on the matching algorithm.

10. Suggested Tech Stack

Layer	Technology
Frontend	Next.js (React Framework)
Backend	Node.js + Express
Database	PostgreSQL
Authentication	JWT (JSON Web Tokens) + Bcrypt
File Storage	AWS S3 / Cloudinary (for Resumes)
Hosting	AWS / Azure / GCP